



QS-SERIES

PROFESSIONAL LINE ARRAY SYSTEMS

USER MANUAL FOR QS SERIES

QS-50

1×10" Line Array

QS-60

2×10" Line Array

QS-70

1×12" Line Array

QS-80

2×12" Line Array

www.louismartinaudio.it | contact@louismartinaudio.it

© Louis Martin Audio – All Rights Reserved

TABLE OF CONTENTS

1.	Introduction & Safety Instructions	
2.	QS-50 — 1×10" Line Array Speaker	
2.1	Overview & Key Features	
2.2	Technical Specifications	
2.3	Cabinet Views & Dimensions	
3.	QS-60 — 2×10" Line Array Speaker	
3.1	Overview & Key Features	
3.2	Technical Specifications	
3.3	Cabinet Views & Dimensions	
4.	QS-70 — 1×12" Line Array Speaker	
4.1	Overview & Key Features	
4.2	Technical Specifications	
4.3	Cabinet Views & Dimensions	
5.	QS-80 — 2×12" Line Array Speaker	

5.1 Overview & Key Features

5.2 Technical Specifications

5.3 Cabinet Views & Dimensions

6. Comparative Specifications

.....

7. Warranty & Service

.....

1. INTRODUCTION & SAFETY INSTRUCTIONS

About the QS-Series

The Louis Martin QS-Series represents the pinnacle of professional line array technology, engineered for the most demanding applications in touring, rental, and fixed installation markets. Each model in the series is constructed from high-density, multi-layer birch plywood, finished with a touring-grade textured Polyurea black paint for maximum durability and acoustic integrity. All models feature an integrated rigging system with stainless steel push-pins for safe and efficient deployment.

The QS-Series supports both passive and bi-amplified (bi-amp) operating modes, giving system engineers the flexibility to optimize performance for any application. Whether deployed for concerts, live events, theaters, houses of worship, or outdoor venues, QS-Series speakers deliver consistent, powerful, and pristine sound reinforcement.

■ Safety Instructions

Flying & Rigging

When flying speakers, appropriate hardware and techniques must be employed for safely suspending the enclosures. All speaker hardware and flying apparatus must be regularly inspected for any visual or mechanical failure prior to each use.

Audience Safety

Always leave adequate distance between speakers and the audience.
Refer to your local authority for Health and Safety guidance when using loudspeaker systems.

Digital Controller

The Louis Martin digital controller must only be programmed by an authorized technician.
Do not use Louis Martin loudspeaker systems without the correct controller and program.
If a system fails due to incorrect controller use, the warranty is void.

Environmental Protection

Protect your speakers and electronics from exposure to humidity, water, or rain at all times. Do not operate equipment in wet or excessively humid conditions.

Maximum Array Size

Observe the maximum number of boxes per line as specified for each model. Exceeding these limits can result in structural failure and serious injury.

2. QS-50 1×10" LINE ARRAY SPEAKER



Figure 2.1 — QS-50 Product View (Source image: 748 × 460 px)

2.1 Overview

The QS-50 is carefully designed and engineered to meet the rigorous demands of touring performances, rental services, and fixed installations. This bi-amplified and passive switchable model features one 10-inch neodymium low-frequency driver and two 1.4-inch high-frequency compression drivers, ensuring exceptional sound quality and performance.

Constructed from birch plywood, the QS-50 ensures longevity and great acoustic properties for excellent vocal articulation and clarity even at full load. It provides a horizontal coverage of 100°, allowing for optimal sound distribution in various environments. The integral rigging system facilitates easy installation and modular configuration, making it an ideal choice for small touring and fixed installation applications.

Typical Applications:

- | | |
|---|--|
| • Concerts & Live Events • Theatres & Performing Arts •
Outdoor Venues | • Small-to-Medium Venues • Houses of Worship •
Corporate AV |
|---|--|

Key Features:

✓ For Touring & Fixed Installations	✓ 500W RMS Power in Passive Mode
✓ Crossover Modes: Passive / Bi-Amp Switchable	✓ Up to 12 Boxes per line (safety standard)
✓ 1 × 10" Neodymium LF Driver, 3" Voice Coil	✓ 2 × 1.75" Neodymium HF Compression Driver
✓ Integrated rigging system with stainless steel push-pins	✓ Touring grade textured Polyurea black paint finish

2.2 Technical Specifications

Type	Two-way full range Bi-amplified/passive switchable Line Array Speaker
Frequency Response (-10 dB)	65 Hz – 20 kHz
LF Drivers	1 × 10" (254 mm) Neodymium driver with 3" voice coil
HF Drivers	2 × 1.75" (44 mm) HF Compression Driver
Power — Passive Mode	500W (RMS) / 1000W (Program) / 2000W (Peak)
Power — Bi-amp Mode (RMS)	LF: 350W HF: 150W
Power — Bi-amp Mode (Program)	LF: 700W HF: 300W
Power — Bi-amp Mode (Peak)	LF: 1400W HF: 600W
Sensitivity (1W/1m)	101 dB
Max SPL — Continuous / Peak (1m)	131 dB / 134 dB
Nominal Impedance	8 Ohms (Passive / Bi-Amp)
Crossover Mode	Passive / Bi-amp Switchable
Crossover Point — LF (Bi-amp)	65 Hz – 2.2 kHz
Crossover Point — HF (Bi-amp)	2.2 kHz – 20 kHz
Dispersion	100° (H) × 10° (V)
Cabinet Material	High density multi-layer birch plywood with Polyurea Paint
Dimensions (H × W × D)	286 mm × 478 mm × 360 mm
Weight	16.8 kg / 37 lbs
Max Boxes Per Line	12

2.3 Cabinet Views & Image Dimensions

The following four-view diagram provides detailed visual reference for the cabinet from all orientations. Image source dimensions are noted for reference.



Side View

Source: 377 × 254 px



Front View

Source: 436 × 313 px



Rear View

Source: 539 × 321 px



Top View

Source: 520 × 321 px



QS-50 Array Stack Configuration

The image shows a typical multi-box array configuration. Always ensure the rigging hardware is rated for the combined weight of all boxes in the array. Maximum 12 boxes per line are permitted under safety standards.

Source image dimensions: 560 × 853 px (rendered to fit)

3. QS-60 2×10" LINE ARRAY SPEAKER

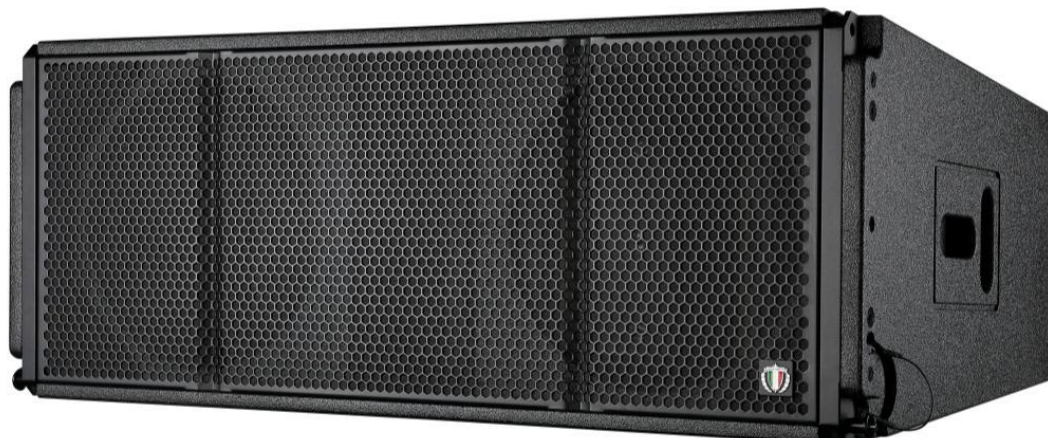


Figure 3.1 — QS-60 Product View (Source image: 1075 × 452 px)

3.1 Overview

The QS-60 is a precision-engineered line array speaker system designed to meet the high-performance demands of touring productions, rental operations, and fixed installations. Built for versatility, durability, and sonic excellence, it delivers consistent, powerful sound reinforcement across a wide range of applications. Its bi-amplified and passive switchable design offers flexible configuration for diverse audio setups.

Equipped with two premium 10-inch Neodymium low-frequency drivers and two 1.75" compression high-frequency drivers, the QS-60 provides a horizontal dispersion of 100°, ensuring consistent sound distribution across various venue types. Constructed from high-density, multi-layer birch plywood, the enclosure prioritizes vocal intelligibility, sonic fidelity, and low distortion, while maintaining a responsive transient profile.

Typical Applications:

- | | |
|---|--|
| • Small-to-Mid Size Events • Outdoor Venues • Houses of Worship | • Theatres & Auditoriums • Corporate Events • Rental Productions |
|---|--|

Key Features:

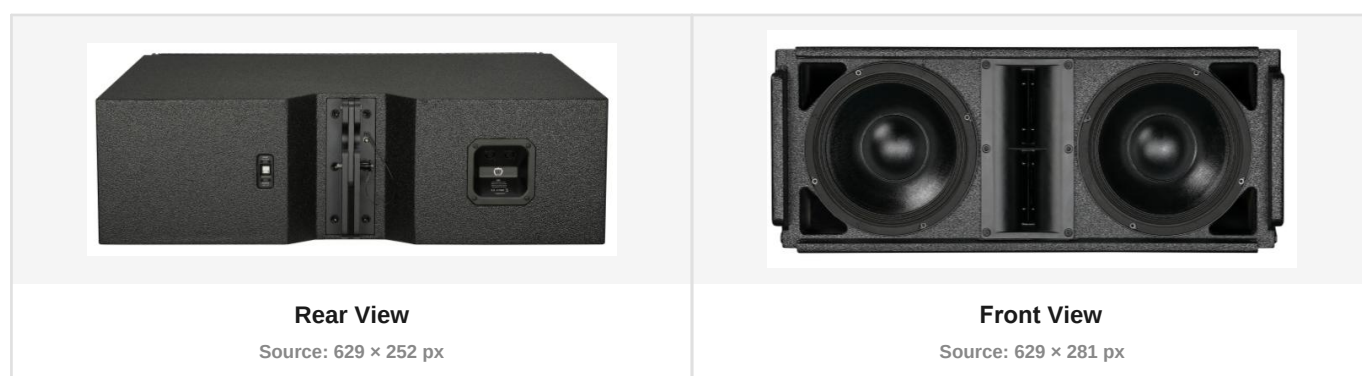
✓ For Touring & Fixed Installations	✓ 750W RMS Power in Passive Mode
✓ Crossover Modes: Passive / Bi-Amp Switchable	✓ Up to 16 Boxes per line (safety standard)
✓ 2 × 10" Neodymium LF Driver, 3" Voice Coil	✓ 2 × 1.75" Neodymium HF Compression Driver
✓ Integrated rigging system with stainless steel push-pins	✓ Touring grade textured Polyurea black paint finish

3.2 Technical Specifications

Type	Two-way full range Bi-amplified/passive switchable Line Array Speaker
Frequency Response (-10 dB)	60 Hz – 20 kHz
LF Drivers	2 × 10" (254 mm) Neodymium driver with 3" voice coil
HF Drivers	2 × 1.75" (44 mm) HF Compression Driver
Power — Passive Mode	750W (RMS) / 1500W (Program) / 3000W (Peak)
Power — Bi-amp Mode (RMS)	LF: 600W HF: 150W
Power — Bi-amp Mode (Program)	LF: 1200W HF: 300W
Power — Bi-amp Mode (Peak)	LF: 2400W HF: 600W
Sensitivity (1W/1m)	102 dB
Max SPL — Continuous / Peak (1m)	133 dB / 136 dB
Nominal Impedance	8 Ohms (Passive / Bi-Amp)
Crossover Mode	Passive / Bi-amp Switchable
Crossover Point — LF (Bi-amp)	60 Hz – 2.2 kHz
Crossover Point — HF (Bi-amp)	2.2 kHz – 20 kHz
Dispersion	100° (H) × 10° (V)
Cabinet Material	High density multi-layer birch plywood with Polyurea Paint
Dimensions (H × W × D)	286 mm × 754 mm × 380 mm
Weight	22.8 kg / 50.2 lbs
Max Boxes Per Line	16

3.3 Cabinet Views & Image Dimensions

The following four-view diagram provides detailed visual reference for the cabinet from all orientations. Image source dimensions are noted for reference.





Side View

Source: 395 × 266 px



Top View

Source: 497 × 266 px



QS-60 Array Stack Configuration

The image shows a typical multi-box array configuration. Always ensure the rigging hardware is rated for the combined weight of all boxes in the array. Maximum 16 boxes per line are permitted under safety standards.

Source image dimensions: 515 × 813 px (rendered to fit)

4. QS-70 1×12" LINE ARRAY SPEAKER



Figure 4.1 — QS-70 Product View (Source image: 748 × 460 px)

4.1 Overview

The QS-70 is carefully designed and engineered to meet the rigorous demands of touring performances, rental services, and fixed installations. This bi-amplified and passive switchable model features one 12-inch neodymium low-frequency driver and two 1.75-inch high-frequency compression driver, ensuring exceptional sound quality and performance with enhanced low-frequency extension compared to the QS-70.

Constructed from birch plywood, the QS-70 ensures longevity and great acoustic properties for excellent vocal articulation and clarity even at full load. It provides a horizontal coverage of 100°, allowing for optimal sound distribution across various environments. The integral rigging system facilitates easy installation and modular configuration, making it an ideal choice for both touring and fixed installations.

Typical Applications:

- Concerts & Live Events • Theatres & Performing Arts • Outdoor Venues
- Small-to-Medium Venues • Houses of Worship • Corporate Audio Visual Events

Key Features:

✓ For Touring & Fixed Installations	✓ 600W RMS Power in Passive Mode
✓ Crossover Modes: Passive / Bi-Amp Switchable	✓ Up to 12 Boxes per line (safety standard)
✓ 1 × 12" Neodymium LF Driver, 3" Voice Coil	✓ 2 × 1.75" Neodymium HF Compression Driver
✓ Integrated rigging system with stainless steel push-pins	✓ Touring grade textured Polyurea black paint finish

4.2 Technical Specifications

Type	Two-way full range Bi-amplified/passive switchable Line Array Speaker
Frequency Response (-10 dB)	60 Hz – 20 kHz
LF Drivers	1 × 12" (304 mm) Neodymium driver with 3" voice coil
HF Drivers	2 × 1.75" (44 mm) HF Compression Driver
Power — Passive Mode	600W (RMS) / 1200W (Program) / 2400W (Peak)
Power — Bi-amp Mode (RMS)	LF: 450W HF: 150W
Power — Bi-amp Mode (Program)	LF: 900W HF: 300W
Power — Bi-amp Mode (Peak)	LF: 1800W HF: 600W
Sensitivity (1W/1m)	103 dB
Max SPL — Continuous / Peak (1m)	133 dB / 136 dB
Nominal Impedance	8 Ohms (Passive / Bi-Amp)
Crossover Mode	Passive / Bi-amp Switchable
Crossover Point — LF (Bi-amp)	60 Hz – 2.2 kHz
Crossover Point — HF (Bi-amp)	2.2 kHz – 20 kHz
Dispersion	100° (H) × 10° (V)
Cabinet Material	High density multi-layer birch plywood with Polyurea Paint
Dimensions (H × W × D)	348 mm × 600 mm × 422 mm
Weight	22 kg / 48.5 lbs
Max Boxes Per Line	12

4.3 Cabinet Views & Image Dimensions

The following four-view diagram provides detailed visual reference for the cabinet from all orientations. Image source dimensions are noted for reference.



Side View

Source: 377 × 254 px



Front View

Source: 436 × 313 px



Rear View

Source: 539 × 321 px



Top View

Source: 520 × 321 px



QS-70 Array Stack Configuration

The image shows a typical multi-box array configuration. Always ensure the rigging hardware is rated for the combined weight of all boxes in the array. Maximum 12 boxes per line are permitted under safety standards.

Source image dimensions: 560 × 853 px (rendered to fit)

5. QS-80 2×12" LINE ARRAY SPEAKER

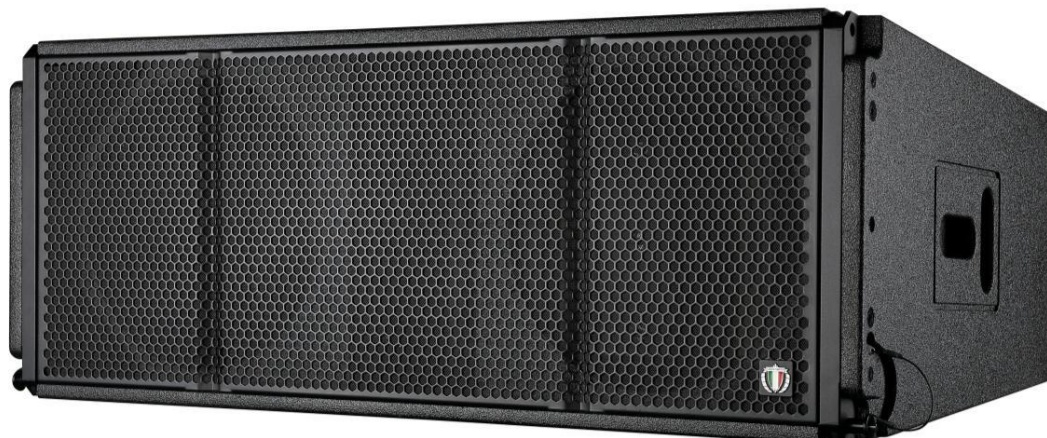


Figure 5.1 — QS-80 Product View (Source image: 1219 × 513 px)

5.1 Overview

The QS-80 is a precision-engineered flagship line array speaker system designed to meet the highest performance demands of major touring productions, rental operations, and large fixed installations. Built for versatility, durability, and sonic excellence, it delivers consistent, powerful sound reinforcement across the widest range of applications. Its bi-amplified and passive switchable design offers maximum configuration flexibility.

Equipped with two premium 12-inch Neodymium low-frequency drivers and two 3-inch compression high-frequency drivers with 1.5-inch exits, the QS-80 provides a horizontal dispersion of 100° with the deepest bass extension in the series (50 Hz). Constructed from high-density, multi-layer birch plywood, the enclosure prioritizes vocal intelligibility, sonic fidelity, and low distortion, while maintaining a responsive transient profile at up to 140 dB peak SPL.

Typical Applications:

- Major Concerts & Live Events • Large Outdoor Venues • Houses of Worship
- Theatres & Auditoriums • Stadium & Arena Events • Large-Scale Rental Productions

Key Features:

✓ For Touring & Fixed Installations	✓ 1200W RMS Power in Passive Mode
✓ Crossover Modes: Passive / Bi-Amp Switchable	✓ Up to 16 Boxes per line (safety standard)
✓ 2 × 12" Neodymium LF Driver, 3" Voice Coil	✓ 2 × 3" Neodymium HF Compression Driver with 1.5" Exit
✓ Integrated rigging system with stainless steel push-pins	✓ Touring grade textured Polyurea black paint finish

5.2 Technical Specifications

Type	Two-way full range Bi-amplified/passive switchable Line Array Speaker
Frequency Response (-10 dB)	50 Hz – 20 kHz
LF Drivers	2 × 12" (304 mm) Neodymium driver with 3" voice coil
HF Drivers	2 × 3" (75 mm) HF Compression Driver with 1.5" Exit
Power — Passive Mode	1200W (RMS) / 2400W (Program) / 4800W (Peak)
Power — Bi-amp Mode (RMS)	LF: 900W HF: 300W
Power — Bi-amp Mode (Program)	LF: 1800W HF: 600W
Power — Bi-amp Mode (Peak)	LF: 3600W HF: 1200W
Sensitivity (1W/1m)	103 dB
Max SPL — Continuous / Peak (1m)	137 dB / 140 dB
Nominal Impedance	8 Ohms (Passive / Bi-Amp)
Crossover Mode	Passive / Bi-amp Switchable
Crossover Point — LF (Bi-amp)	50 Hz – 1.8 kHz
Crossover Point — HF (Bi-amp)	1.8 kHz – 20 kHz
Dispersion	100° (H) × 10° (V)
Cabinet Material	High density multi-layer birch plywood with Polyurea Paint
Dimensions (H × W × D)	348 mm × 910 mm × 520 mm
Weight	36.8 kg / 81.1 lbs
Max Boxes Per Line	16

5.3 Cabinet Views & Image Dimensions

The following four-view diagram provides detailed visual reference for the cabinet from all orientations. Image source dimensions are noted for reference.



Rear View

Source: 666 × 267 px



Top View

Source: 499 × 267 px



Front View

Source: 719 × 302 px

Side View

Source: 408 × 275 px



QS-80 Array Stack Configuration

The image shows a typical multi-box array configuration. Always ensure the rigging hardware is rated for the combined weight of all boxes in the array. Maximum 16 boxes per line are permitted under safety standards.

Source image dimensions: 578 × 913 px (rendered to fit)

6. COMPARATIVE SPECIFICATIONS

The table below provides a side-by-side comparison of all four QS-Series models to assist in selecting the right product for your application.

Parameter	QS-50	QS-60	QS-70	QS-80
Configuration	1×10"	2×10"	1×12"	2×12"
LF Voice Coil	3"	3"	3"	3"
HF Driver	2×1.75"	2×1.75"	2×1.75"	2×3"
Freq. Response (–10dB)	65Hz–20kHz	60Hz–20kHz	60Hz–20kHz	50Hz–20kHz
Passive RMS Power	500W	750W	600W	1200W
Passive Program Power	1000W	1500W	1200W	2400W
Passive Peak Power	2000W	3000W	2400W	4800W
Sensitivity (1W/1m)	101 dB	102 dB	103 dB	103 dB
Max SPL Continuous	131 dB	133 dB	133 dB	137 dB
Max SPL Peak	134 dB	136 dB	136 dB	140 dB
Impedance	8Ω	8Ω	8Ω	8Ω
Dispersion (H×V)	100°×10°	100°×10°	100°×10°	100°×10°
Crossover Point	2.2 kHz	2.2 kHz	2.2 kHz	1.8 kHz
Height (mm)	286	286	348	348
Width (mm)	478	754	600	910
Depth (mm)	360	380	422	520
Weight (kg)	16.8	22.8	22.0	36.8
Weight (lbs)	37	50.2	48.5	81.1
Max Boxes / Line	12	16	12	16

7. WARRANTY & SERVICE

Warranty Coverage

The loudspeakers and electronics are covered against defects in workmanship or materials for a period of two (2) years from the original date of purchase. This warranty is non-transferable and applies only to the original purchaser.

Warranty Exclusions

The warranty does not cover any item which, in Louis Martin's opinion, has failed due to user abuse, accident, unauthorized modifications, or incorrect use of the digital controller. Using the system without the correct Louis Martin controller program will void the warranty.

Warranty Service

Defective items will be repaired or replaced with no charge for materials or labour. However, the item must be adequately packed and dispatched, pre-paid, to a Louis Martin authorized distributor or service centre.

Contact

For warranty claims, service inquiries, or technical support, please contact Louis Martin Audio at contact@louismartinaudio.it or visit www.louismartinaudio.it.

www.louismartinaudio.it | contact@louismartinaudio.it

© Louis Martin Audio – All Rights Reserved